

A REFINEMENT OF THE FIXED–PIXED POINTS EQUIDISTRIBUTION ON RESTRICTED PERMUTATIONS

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ABSTRACT. Motivated by a recent conjecture of Bsila, Cox, Hugo, Styron and Zhuang concerning fixed points and pixed points on pattern-avoiding permutations, we prove a bivariate refinement involving descent statistics. Given a set of permutations Π , let $\mathfrak{S}_n(\Pi)$ denote the set of permutations in the symmetric group $\mathfrak{S}_n$ that avoid every element of Π in the sense of pattern avoidance. For each set Π appearing in their conjecture, we show that the pairs of statistics (des, fix) and $(\text{ides}, \text{pix})$ are equidistributed over $\mathfrak{S}_n(\Pi)$. Our proof is based on explicit ordinary generating functions for the corresponding pattern-avoiding classes.

1. INTRODUCTION

Let $\mathfrak{S}_n$ denote the symmetric group of permutations of the set $[n] = \{1, 2, \dots, n\}$. We call $i \in [n]$ a *fixed point* of $\sigma \in \mathfrak{S}_n$ if $\sigma(i) = i$. Let $\text{fix}(\sigma)$ denote the number of fixed points of σ . A permutation $\sigma \in \mathfrak{S}_n$ is called a *derangement* if $\text{fix}(\sigma) = 0$. Researches on fixed points and derangements can refer to References [Bsi+25; DW93; KZ01; HX09; Lin16; Eli12]. We call i an *ascent* of σ if $\sigma(i) < \sigma(i+1)$ for $i \in [n-1]$ or $i = n$. In the 1980s, Désarménien introduced another family of permutations, now called *desarrangements*, and proved that they are equinumerous with derangements [Dés84]. A desarrangement is a permutation whose first ascent is even. Note that a decreasing permutation $nn-1 \cdots 1$ is a desarrangement if and only if n is even. The empty word is also regarded as a desarrangement.

Example 1. For $1 \leq n \leq 4$, the desarrangements in $\mathfrak{S}_n$ are, respectively,

$$\emptyset, \quad \{21\}, \quad \{213, 312\},$$

and

$$\{2134, 2143, 3124, 3142, 3241, 4123, 4132, 4231, 4321\}.$$

The intrinsic relationship between derangements and desarrangements has been further explored on the basis of the definition of pixed points. As a combinatorial statistic on permutations, pixed points were originally proposed and thoroughly studied by Foata and Han in the setting of hyperoctahedral groups. Foata and Han [FH08] introduced the pixed factorization of a permutation, that is, every permutation $\sigma \in \mathfrak{S}_n$ admits a unique factorization $\sigma = \iota\delta$, where ι is an increasing prefix and δ is a desarrangement. The letters of ι are the *pixed points* of σ , and we denote by $\text{pix}(\sigma)$ the number of pixed points of σ . In particular, the identity permutation has n pixed points. Under this terminology, derangements are the permutations with no fixed points, while desarrangements are the permutations with no pixed points.

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Example 2. Let $\sigma = 135764829 \in \mathfrak{S}_9$. Then the *pixed* factorization of σ is

$$\sigma = \underbrace{1357}_{\iota} \underbrace{64829}_{\delta}.$$

Here $\iota = 1357$ is the longest increasing prefix of σ and the first ascent of $\delta = 64829$ occurs at position 2. Thus δ is a *desarrangement* and $\text{pix}(\sigma) = 4$.

Pattern avoidance is a central topic in the modern study of permutations. Given two permutations $\sigma \in \mathfrak{S}_n$ and $\tau \in \mathfrak{S}_k$, we say that σ contains the pattern τ if there exist indices $1 \leq i_1 < i_2 < \cdots < i_k \leq n$ such that the subsequence $\sigma(i_1)\sigma(i_2)\cdots\sigma(i_k)$ has the same relative order as τ . Otherwise, we say that σ *avoids* τ . For example, the permutation $\sigma = 31524$ contains the pattern 132, since the subsequence 354 has the same relative order as 132. On the other hand, $\sigma = 31524$ avoids pattern 321, since it has no decreasing subsequence of length 3. More generally, for a set of patterns Π , we write

$$\mathfrak{S}_n(\Pi) = \{\sigma \in \mathfrak{S}_n : \sigma \text{ avoids every pattern in } \Pi\}. \quad (1.1)$$

When the patterns in Π are listed explicitly, we often omit the enclosing braces. For example, $\mathfrak{S}_n(123, 132)$ stands for $\mathfrak{S}_n(\{123, 132\})$.

The study of pattern-avoiding permutations goes back to the work of Knuth on stack-sortable permutations, where the permutations sortable by a single stack were shown to be precisely the 231-avoiding permutations [Knu98]. These permutations are counted by the Catalan numbers, and this observation initiated a rich interaction between permutation patterns and classical Catalan objects. Later, Simion and Schmidt [SS85] carried out a systematic study of permutations avoiding patterns of length three, proving in particular that all single patterns of length three are Catalan-enumerated. Over the past few decades, pattern avoidance has become a broad framework for studying refined enumeration, bijective constructions, permutation statistics, and structural decompositions of permutation classes; see [Kit11; MR02] for an overview.

Recently, Bsila, Cox, Hugo, Styron and Zhuang revisited desarrangements from the point of view of permutation statistics and pattern avoidance [Bsi+25]. They obtained several generating functions for desarrangements with respect to some permutation statistics, including descents, peaks, valleys, double ascents and double descents, and they also gave a complete enumeration of desarrangements avoiding prescribed sets of patterns of length three. Their work led to new interpretations of several classical integer sequences, including Catalan, Fine, Jacobsthal and Fibonacci numbers, in terms of pattern-avoiding desarrangements.

Let

$$\mathcal{P} = \left\{ \begin{array}{lll} \{132, 312\}, & \{132, 321\}, & \{213, 231\}, \\ \{123, 132, 312\}, & \{123, 213, 231\}, & \{123, 312, 321\}, \\ \{132, 312, 321\}, & \{213, 231, 312\}, & \{213, 231, 321\} \end{array} \right\}. \quad (1.2)$$

At the end of their paper, Bsila et al. [Bsi+25] posed the following conjecture ¹.

Conjecture 1.1. For all $n \geq 0$ and for every $\Pi \in \mathcal{P}$, the permutation statistics *fix* and *pix* are equidistributed over $\mathfrak{S}_n(\Pi)$. Equivalently,

$$\sum_{\sigma \in \mathfrak{S}_n(\Pi)} x^{\text{fix}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n(\Pi)} x^{\text{pix}(\sigma)}. \quad (1.3)$$

¹After completing our work, we learned that Zhuang and his students had independently obtained a bijective proof of the original conjecture [Zhu26].

The purpose of this paper is to prove a refinement of this conjecture. Instead of considering only the one-variable distributions of fix and pix , we include descent statistics. We call $i \in [n-1]$ a *descent* of $\sigma \in \mathfrak{S}_n$ if $\sigma(i) > \sigma(i+1)$. We denote by $\text{Des}(\sigma)$ the set of descents of σ , and denote its cardinality by $\text{des}(\sigma) = |\text{Des}(\sigma)|$. We call $i \in [n-1]$ an *inverse descent* of $\sigma \in \mathfrak{S}_n$ if $\sigma^{-1}(i) > \sigma^{-1}(i+1)$, and we denote by $\text{ides}(\sigma)$ the number of inverse descents of σ . Equivalently, i is an inverse descent of σ if $i+1$ appears to the left of i in σ . Note that $\text{ides}(\sigma)$ is simply the number of $i \in [n-1]$ such that $i+1$ appears somewhere to the left of i in σ .

Example 3. Let $\sigma = 314265 \in \mathfrak{S}_6$. The descents of σ are 1, 3 and 5, since $3 > 1$, $4 > 2$, and $6 > 5$. Hence $\text{des}(\sigma) = 3$. Moreover, $\sigma^{-1} = 241365$, whose descents are 2 and 5. Equivalently, the inverse descents of σ are obtained by comparing the positions of i and $i+1$: here 3 appears to the left of 2, and 6 appears to the left of 5. Therefore, the inverse descents of σ are 2 and 5, and $\text{ides}(\sigma) = 2$.

For each pattern set $\Pi \in \mathcal{P}$, we define the following two enumerative polynomials

$$F_{\Pi}(t; x, y) := \sum_{n \geq 0} \left(\sum_{\pi \in \mathfrak{S}_n(\Pi)} x^{\text{des}(\pi)} y^{\text{fix}(\pi)} \right) t^n \quad (1.4)$$

and

$$P_{\Pi}(t; x, y) := \sum_{n \geq 0} \left(\sum_{\pi \in \mathfrak{S}_n(\Pi)} x^{\text{ides}(\pi)} y^{\text{pix}(\pi)} \right) t^n. \quad (1.5)$$

The following theorem is our main result.

Theorem 1.2. *For every $\Pi \in \mathcal{P}$, we have*

$$F_{\Pi}(t; x, y) = P_{\Pi}(t; x, y). \quad (1.6)$$

Equivalently, for $n \geq 0$, the two pairs (des, fix) and $(\text{ides}, \text{pix})$ are equidistributed over $\mathfrak{S}_n(\Pi)$.

Remark 1.1. *For a pattern set $\Pi \in \mathcal{P}$, let*

$$d_n(\Pi) := |\{\sigma \in \mathfrak{S}_n(\Pi) : \text{fix}(\sigma) = 0\}| \text{ and } \tilde{d}_n(\Pi) := |\{\sigma \in \mathfrak{S}_n(\Pi) : \text{pix}(\sigma) = 0\}|. \quad (1.7)$$

Thus $d_n(\Pi)$ (resp. $\tilde{d}_n(\Pi)$) is the number of derangements (resp. desarrangements) in $\mathfrak{S}_n(\Pi)$. It follows from setting $x = 1$ and $y = 0$ in Theorem 1.2 that

$$d_n(\Pi) = \tilde{d}_n(\Pi), \quad (1.8)$$

for all $n \geq 0$ and all $\Pi \in \mathcal{P}$. The numbers $d_n(\Pi)$ can be obtained from the study of enumerating the fixed point refinement of pattern-avoiding permutations by Robertson–Saracino–Zeilberger [RSZ02] and Mansour–Robertson [MR02]. Theorem 1.2 refines the nine pattern classes belonging to $\mathcal{P}$ in Theorem 3.28 of Bsi et al. [Bsi+25]. Note that the additional case $\Pi = \{132\}$ appearing in their theorem is not covered by this refinement, since it does not satisfy the full equidistribution of fix and pix .

We shall prove Theorem 1.2 by generating function techniques. The two-pattern and three-pattern cases are treated in Sections 2.1 and 2.2, respectively.

2. PROOF OF THEOREM 1.2

Before proving Theorem 1.2, we recall two basic operators on permutations. For any $\sigma \in \mathfrak{S}_n$, its *reversal* $\sigma^r \in \mathfrak{S}_n$ is given by $\sigma^r(i) = \sigma(n+1-i)$; its *complement* $\sigma^c \in \mathfrak{S}_n$ is given by $\sigma^c(i) = n+1-\sigma(i)$. The *reversal-complement* of σ is defined by

$$\sigma^{rc} := (\sigma^r)^c = (\sigma^c)^r. \quad (2.1)$$

Equivalently,

$$\sigma^{rc}(i) = n+1-\sigma(n+1-i) \text{ for } 1 \leq i \leq n. \quad (2.2)$$

In what follows, we shall use the following elementary fact repeatedly.

Lemma 2.1. *The reverse-complement map $\sigma \mapsto \sigma^{rc}$ preserves the statistics des and fix . For any $\sigma \in \mathfrak{S}_n$, we have*

$$\text{des}(\sigma^{rc}) = \text{des}(\sigma) \text{ and } \text{fix}(\sigma^{rc}) = \text{fix}(\sigma). \quad (2.3)$$

Proof. For $1 \leq i \leq n-1$, by the definition of σ^{rc} , we have

$$\begin{aligned} i \in \text{Des}(\sigma^{rc}) &\iff \sigma^{rc}(i) > \sigma^{rc}(i+1) \\ &\iff n+1-\sigma(n+1-i) > n+1-\sigma(n-i) \\ &\iff \sigma(n+1-i) < \sigma(n-i) \\ &\iff n-i \in \text{Des}(\sigma), \end{aligned}$$

and for $1 \leq i \leq n$,

$$\begin{aligned} \sigma^{rc}(i) = i &\iff n+1-\sigma(n+1-i) = i \\ &\iff \sigma(n+1-i) = n+1-i. \end{aligned}$$

It follows that $\sigma \mapsto \sigma^{rc}$ preserves both des and fix . □

2.1. Double-avoidance classes in $\mathcal{P}$. In this section, we prove Theorem 1.2 for three double-avoiding classes Π , namely $\{132, 312\}$, $\{213, 231\}$ and $\{132, 321\}$, by computing their ordinary generating functions $F_\Pi(t; x, y)$ and $P_\Pi(t; x, y)$, respectively.

2.1.1. The class $\mathfrak{S}_n(132, 312)$. Given a permutation $\sigma \in \mathfrak{S}_n$, we call $i \in \{2, 3, \dots, n\}$ an *ascent top position* (resp. a *descent bottom position*) if $\sigma(i-1) < \sigma(i)$ (resp. $\sigma(i-1) > \sigma(i)$). Let $\text{Asct}(\sigma)$ be the set of all ascent tops of σ . A well-established bijection ξ between the restricted permutation class $\mathfrak{S}_n(132, 312)$ and the power set $2^{\{2, 3, \dots, n\}}$ originally introduced by Simion and Schmidt [SS85], is defined by

$$\sigma \longmapsto \xi(\sigma) := \text{Asct}(\sigma). \quad (2.4)$$

Inversely, for any given subset $A = \{a_1 < a_2 < \dots < a_m\} \in 2^{\{2, 3, \dots, n\}}$, the corresponding permutation σ is uniquely constructed by setting

$$\sigma(a_i) = n - m + i \text{ for } 1 \leq i \leq m, \quad (2.5)$$

while the remaining positions are filled, from left to right, by $n-m, n-m-1, \dots, 1$ in strictly decreasing order. The following lemma shows how this bijection provides a natural way to evaluate the permutation statistics des and ides on $\mathfrak{S}_n(132, 312)$.

Lemma 2.2. *Let $\sigma \in \mathfrak{S}_n(132, 312)$, and set*

$$B = [n] \setminus \text{Asct}(\sigma). \quad (2.6)$$

Then we have

$$\text{des}(\sigma) = \text{idcs}(\sigma) = |B| - 1. \quad (2.7)$$

Proof. We first compute $\text{des}(\sigma)$. Since every index $i \in \{2, \dots, n\}$ is either an ascent top position or a descent bottom position of σ , we see that indices in $\{2, \dots, n\} \setminus \text{Asct}(\sigma)$ are exactly the descent bottoms of σ . Then, by (2.6), $\text{des}(\sigma) = |B| - 1$.

It remains to compute $\text{idcs}(\sigma)$. Set $m = |\text{Asct}(\sigma)|$. If $m = 0$, then $\sigma = n(n-1)\cdots 1$, and hence $\text{idcs}(\sigma) = n-1 = |B| - 1$. Thus, assume $m \geq 1$. By (2.5), the permutation σ is constructed by placing the m largest values $\{n-m+1, \dots, n\}$ in strictly increasing order at positions in $\text{Asct}(\sigma)$, and the remaining $n-m$ values $\{1, \dots, n-m\}$ in strictly decreasing order at positions in B . An inverse descent occurs for a value v if and only if $v+1$ appears to the left of v in σ . One can check that

- For $v \in \{n-m+1, \dots, n-1\}$: These values are placed in increasing order, so $v+1$ is always to the right of v . Hence, there are no inverse descents.
- For $v \in \{1, \dots, n-m-1\}$: These values are placed in strictly decreasing order from left to right. Thus, the larger value $v+1$ is always placed before (to the left of) v . Hence, there are exactly $n-m-1$ inverse descents.
- For $v = n-m$: Since $1 \notin \text{Asct}(\sigma)$, position 1 belongs to B , giving $\sigma(1) = n-m$. The value $v+1 = n-m+1$ is placed at the first ascent position $a_1 \geq 2$. Thus, v is to the left of $v+1$, which means that v is not an inverse descent.

It follows from the above argument that $\text{idcs}(\sigma) = |B| - 1$. $\square$

Example 4. Let $\sigma = 67548931021 \in \mathfrak{S}_{10}$. On one hand, we have $\text{Asct}(\sigma) = \{2, 5, 6, 8\}$ and $B = [10] \setminus \text{Asct}(\sigma) = \{1, 3, 4, 7, 9, 10\}$. On the other hand, we have $\text{des}(\sigma) = |\{2, 3, 6, 8, 9\}|$ and $\text{idcs}(\sigma) = |\{1, 2, 3, 4, 5\}|$. Thus $\text{des}(\sigma) = \text{idcs}(\sigma) = |B| - 1$.

Theorem 2.3. *Let $T_1 = \{132, 312\}$. We have*

$$F_{T_1}(t; x, y) = 1 + \frac{yt}{1-yt} + \frac{xt^2(1+xyt)(1-xt)}{(1-yt)(1-x^2t^2)(1-(1+x)t)}. \quad (2.8)$$

Proof. Suppose that $B = \{b_1 < b_2 < \dots < b_N\}$. Under the reverse construction of the Simion–Schmidt bijection ξ defined by (2.5), we have $\sigma(b_j) = N+1-j$ for $1 \leq j \leq N$, i.e., the entries in the positions belonging to B are filled, from left to right, by $N, N-1, \dots, 1$. It follows that b_j is a fixed point if and only if

$$b_j = N+1-j, \quad (2.9)$$

or equivalently,

$$b_j + j = N+1. \quad (2.10)$$

Since the sequence $b_j + j$ is strictly increasing, this can happen for at most one index j . Define

$$\epsilon(B) = \begin{cases} 1, & \text{if there exists } j \text{ such that } b_j + j = N+1; \\ 0, & \text{otherwise.} \end{cases} \quad (2.11)$$

Since b_N is the maximum element of the set B , it follows that all positions $b_N + 1, b_N + 2, \dots, n$ are contained in $\text{Asct}(\sigma)$. It is easy to see that by the reverse construction of ξ , these positions are forced to be fixed points. It follows that

$$\text{fix}(\sigma) = n - b_N + \epsilon(B). \quad (2.12)$$

By combining the bijection ξ with Lemma 2.2 and (2.12), the polynomials $F_\Pi(t; x, y)$ (for $\Pi = T_1 = \{132, 312\}$) defined in (1.4) can be reformulated in the following manner,

$$F_{T_1}(t; x, y) = 1 + \sum_{n \geq 1} \sum_{\substack{B \subseteq [n] \\ 1 \in B}} x^{|B|-1} y^{n - \max B + \epsilon(B)} t^n, \quad (2.13)$$

where $\max B$ denotes the maximum element of set B . Let $\mathbb{P}$ be the set of positive integers. By first fixing B and summing over all $n \geq \max B$, we obtain

$$\begin{aligned} F_{T_1}(t; x, y) &= 1 + \sum_{\substack{B \subseteq \mathbb{P} \\ 1 \in B}} x^{|B|-1} y^{\epsilon(B)} \sum_{n \geq \max B} y^{n - \max B} t^n \\ &= 1 + \frac{1}{1 - yt} \sum_{\substack{B \subseteq \mathbb{P} \\ 1 \in B}} x^{|B|-1} y^{\epsilon(B)} t^{\max B}. \end{aligned} \quad (2.14)$$

Define the enumerative polynomials

$$H(t; x, y) := \sum_{\substack{B \subseteq \mathbb{P} \\ 1 \in B}} x^{|B|-1} y^{\epsilon(B)} t^{\max B}. \quad (2.15)$$

We now compute $H(t; x, y)$. First ignore the statistic $\epsilon(B)$, and set

$$H_0(t; x) := \sum_{\substack{B \subseteq \mathbb{P} \\ 1 \in B}} x^{|B|-1} t^{\max B}. \quad (2.16)$$

If $B = \{1\}$, the contribution is t . If $\max B = L \geq 2$, then 1 and L are forced to be in B , while the elements of $\{2, 3, \dots, L-1\}$ may be chosen freely. Hence,

$$H_0(t; x) = t + \sum_{L \geq 2} x(1+x)^{L-2} t^L. \quad (2.17)$$

Next, we compute the contribution of those sets B satisfying $\epsilon(B) = 1$. Let

$$H_1(t; x) := \sum_{\substack{B \subseteq \mathbb{P} \\ 1 \in B, \epsilon(B)=1}} x^{|B|-1} t^{\max B}. \quad (2.18)$$

The set $B = \{1\}$ satisfies $\epsilon(B) = 1$ and contributes t . Now suppose that $B = \{b_1 < b_2 < \dots < b_N\}$ has $N \geq 2$ elements and satisfies $\epsilon(B) = 1$. Then there exist a unique index j such that

$$b_j + j = N + 1. \quad (2.19)$$

Set $p = b_j$. Then $N = p + j - 1$. Since $1 = b_1$ and $b_j = p$, the elements of B before p consist of 1, together with $j - 2$ elements chosen from $\{2, 3, \dots, p-1\}$. Thus there are

$$\binom{p-2}{j-2}$$

choices for the part of B before p .

On the other hand, the number of elements of B after p is $N - j = b_j - 1$. If $L = \max B$, then one of these elements is L , and the remaining $p - 2$ elements are chosen from $\{p + 1, p + 2, \dots, L - 1\}$. Thus, for fixed p and L , the number of choices after p is

$$\binom{L - p - 1}{p - 2}.$$

Summing over all possible L , we obtain

$$\sum_{L \geq 2p-1} \binom{L - p - 1}{p - 2} t^L = \frac{t^{2p-1}}{(1 - t)^{p-1}}. \quad (2.20)$$

Moreover, since $N = p + j - 1$, the exponent of x is $|B| - 1 = N - 1 = p + j - 2$. Therefore

$$H_1(t; x) = t + \sum_{p \geq 2} \sum_{j=2}^p \binom{p-2}{j-2} x^{p+j-2} \frac{t^{2p-1}}{(1-t)^{p-1}}. \quad (2.21)$$

Setting $k = p - 2$ and $i = j - 2$, we get

$$H_1(t; x) - t = \sum_{k \geq 0} \frac{x^{k+2} t^{2k+3}}{(1-t)^{k+1}} \sum_{i=0}^k \binom{k}{i} x^i. \quad (2.22)$$

Combining the binomial theorem and the geometric series identity, we derive

$$H_1(t; x) = t + \frac{x^2 t^3}{1 - t - x(1+x)t^2}. \quad (2.23)$$

Since y only records whether the possible extra fixed point inside B exists, we have

$$H(t; x, y) = H_0(t; x) + (y - 1)H_1(t; x). \quad (2.24)$$

Substituting (2.17) and (2.23) into (2.24), and using

$$1 - t - x(1+x)t^2 = (1+xt)(1 - (1+x)t),$$

we obtain

$$H(t; x, y) = yt + \frac{xt^2(1+xyt)(1-xt)}{(1-x^2t^2)(1-(1+x)t)}. \quad (2.25)$$

Combining this with (2.14) and (2.15), we obtain (2.8). $\square$

Theorem 2.4. *Let $T_1 = \{132, 312\}$. We have*

$$P_{T_1}(t; x, y) = 1 + \frac{yt}{1-yt} + \frac{xt^2(1+xyt)(1-xt)}{(1-yt)(1-x^2t^2)(1-(1+x)t)}. \quad (2.26)$$

Proof. We first consider the case $\text{Asct}(\sigma) = \{2, 3, \dots, n\}$. Then $\sigma = 12 \cdots n$ and $B = [n] \setminus \text{Asct}(\sigma) = \{1\}$. Hence, by the pixed factorization, we have $\text{pix}(\sigma) = n$, and by Lemma 2.2, we have $\text{ides}(\sigma) = 0$. Therefore, this case and the empty permutation contribute

$$1 + \sum_{n \geq 1} y^n t^n = 1 + \frac{yt}{1-yt}. \quad (2.27)$$

For the remaining part, assume that $|B| \geq 2$ and $B = \{b_1 < b_2 < \dots < b_N\}$ with $b_1 = 1$. Since $B = [n] \setminus \text{Asct}(\sigma)$, the positions $2, 3, \dots, b_2 - 1$ belong to $\text{Asct}(\sigma)$. Hence the maximal

increasing prefix of σ has length $b_2 - 1$. Now let $r \geq 1$ be the length of the maximal consecutive segment of elements of B beginning at b_2 , i.e.,

$$b_2, b_2 + 1, \dots, b_2 + r - 1 \in B, \quad (2.28)$$

and either $b_2 + r \notin B$ or $b_2 + r > n$. Define

$$\eta(B) = \begin{cases} 0, & \text{if } r \text{ is even;} \\ 1, & \text{if } r \text{ is odd.} \end{cases} \quad (2.29)$$

One can check that the number of fixed points of σ is decided by the parity of r ,

$$\text{pix}(\sigma) = b_2 - 1 - \eta(B), \quad (2.30)$$

or equivalently,

$$\text{pix}(\sigma) = \begin{cases} b_2 - 1, & \text{if } r \text{ is even;} \\ b_2 - 2, & \text{if } r \text{ is odd.} \end{cases} \quad (2.31)$$

By combining the bijection ξ with Lemma 2.2, (2.30) and (2.27), the polynomials $P_\Pi(t; x, y)$ (for $\Pi = T_1 = \{132, 312\}$), defined in (1.5), can be reformulated as follows:

$$P_{T_1}(t; x, y) = 1 + \frac{yt}{1 - yt} + \sum_{n \geq 2} \sum_{\substack{B \subseteq [n] \\ 1 \in B, |B| \geq 2}} x^{|B|-1} y^{b_2-1-\eta(B)} t^n. \quad (2.32)$$

By first fixing B and summing over all $n \geq \max B$, we obtain

$$\begin{aligned} P_{T_1}(t; x, y) &= 1 + \frac{yt}{1 - yt} + \sum_{\substack{B \subseteq [n] \\ 1 \in B, |B| \geq 2}} x^{|B|-1} y^{b_2-1-\eta(B)} \sum_{n \geq \max B} t^n \\ &= 1 + \frac{yt}{1 - yt} + \frac{1}{1 - t} \sum_{\substack{B \subseteq \mathbb{P} \\ 1 \in B, |B| \geq 2}} x^{|B|-1} y^{b_2-1-\eta(B)} t^{\max B}. \end{aligned} \quad (2.33)$$

Define the following enumerative polynomials

$$K(t; x, y) = \sum_{\substack{B \subseteq \mathbb{P} \\ 1 \in B, |B| \geq 2}} x^{|B|-1} y^{b_2-1-\eta(B)} t^{\max B}. \quad (2.34)$$

Combining this with (2.33), we have (for $\Pi = T_1 = \{132, 312\}$)

$$P_\Pi(t; x, y) = 1 + \frac{yt}{1 - yt} + \frac{K(t; x, y)}{1 - t}. \quad (2.35)$$

We now enumerate finite subsets B of positive integers containing 1 and satisfying $|B| \geq 2$. Fix $\ell := b_2 - 1 \geq 1$ and $r \geq 1$. Then B begins as

$$B = \{1\} \cup \{\ell + 1, \ell + 2, \dots, \ell + r\} \cup \{\text{later elements}\}.$$

We now count the possible later elements.

- If there are no later elements, then $\max B = \ell + r$, and the contribution is $t^{\ell+r}$.

- If there are later elements, then $\ell + r + 1 \notin B$, by maximality of the segment of consecutive elements, and the next possible elements lie in

$$\{\ell + r + 2, \ell + r + 3, \dots\}.$$

If the maximum is $L := \max B \geq \ell + r + 2$, then L is forced to belong to B , and the elements of

$$\{\ell + r + 2, \ell + r + 3, \dots, L - 1\}$$

may be chosen freely. Thus the contribution of the later part is

$$\sum_{L \geq \ell + r + 2} x(1+x)^{L-(\ell+r+2)} t^L = \frac{xt^{\ell+r+2}}{1-(1+x)t}.$$

Therefore, for fixed ℓ and r , the contribution of the final part is

$$t^{\ell+r} + \frac{xt^{\ell+r+2}}{1-(1+x)t} = t^{\ell+r} \frac{(1-t)(1-xt)}{1-(1+x)t}. \quad (2.36)$$

Since x records $|B| - 1$, the segment $\ell + 1, \ell + 2, \dots, \ell + r$ contributes x^r . Combining this with (2.36), we derive that $K(t; x, y)$ defined in (2.34) is equal to

$$K(t; x, y) = \frac{(1-t)(1-xt)}{1-(1+x)t} \sum_{\ell \geq 1} \sum_{r \geq 1} x^r t^{\ell+r} y^{\ell-\chi(2 \nmid r)}, \quad (2.37)$$

where $\chi(P)$ is equal to 1 if the proposition P is true, and 0 otherwise. Separating r according to its parity, we get

$$\begin{aligned} K(t; x, y) &= \frac{(1-t)(1-xt)}{1-(1+x)t} \sum_{\ell \geq 1} t^\ell \left(y^\ell \sum_{\substack{r \geq 1 \\ r \text{ even}}} (xt)^r + y^{\ell-1} \sum_{\substack{r \geq 1 \\ r \text{ odd}}} (xt)^r \right) \\ &= \frac{(1-t)(1-xt)}{1-(1+x)t} \sum_{\ell \geq 1} t^\ell \left(y^\ell \frac{x^2 t^2}{1-x^2 t^2} + y^{\ell-1} \frac{xt}{1-x^2 t^2} \right) \\ &= \frac{xt^2(1+xyt)(1-t)(1-xt)}{(1-yt)(1-x^2 t^2)(1-(1+x)t)}. \end{aligned} \quad (2.38)$$

Substituting $K(t; x, y)$ by (2.38) in (2.35), we obtain (2.26). $\square$

2.1.2. The class $\mathfrak{S}_n(213, 231)$. We first describe the structure of permutations in $\mathfrak{S}_n(213, 231)$. Let $\sigma \in \mathfrak{S}_n(213, 231)$, and write

$$\sigma = L n R. \quad (2.39)$$

where L and R are the words to the left and right of the letter n , respectively. Since σ avoids 213, the word L must be increasing. Moreover, since σ avoids 231, every letter in L must be smaller than every letter in R . It follows that

$$L = 12 \cdots a \text{ for some } 0 \leq a \leq n-1. \quad (2.40)$$

More precisely, every $\sigma \in \mathfrak{S}_n(213, 231)$ has a unique decomposition

$$\sigma = 12 \cdots a n \tau^{+a}, \quad (2.41)$$

where τ^{+a} is obtained by adding a to each letter of $\tau \in \mathfrak{S}_{n-a-1}(213, 231)$.

Example 5. Let $\sigma = 1237465 \in \mathfrak{S}_7(213, 231)$. Then the letter 7 separates σ as $\sigma = L7R$, where $L = 123$ and $R = \tau^{+3} = 465$. Thus $a = 3$. Subtracting $a = 3$ from each letter of R , we obtain $\tau = 132 \in \mathfrak{S}_3(213, 231)$.

Theorem 2.5. Let $T_2 = \{213, 231\}$. We have

$$F_{T_2}(t; x, y) = 1 + \frac{yt}{1 - yt} + \frac{xt^2(1 + xyt)(1 - xt)}{(1 - yt)(1 - x^2t^2)(1 - (1 + x)t)}. \quad (2.42)$$

Proof. By Lemma 2.1, the reversal-complement map $\sigma \mapsto \sigma^{rc}$ restricts a bijection between $\mathfrak{S}_n(132, 312)$ and $\mathfrak{S}_n(213, 231)$, thus we have $F_{T_2}(t; x, y) = F_{T_1}(t; x, y)$, where $T_1 = \{132, 312\}$. By the formula for $F_{T_1}(t; x, y)$ given in (2.8), we obtain (2.42). $\square$

Theorem 2.6. Let $T_2 = \{213, 231\}$. We have

$$P_{T_2}(t; x, y) = 1 + \frac{yt}{1 - yt} + \frac{xt^2(1 + xyt)(1 - xt)}{(1 - yt)(1 - x^2t^2)(1 - (1 + x)t)}. \quad (2.43)$$

Proof. From the decomposition (2.41), if $a = n - 1$, then $\sigma = 12 \cdots n$. Hence, $\text{ides}(\sigma) = 0$ and $\text{pix}(\sigma) = n$. Together with the empty permutation, these identity permutations contribute

$$1 + \sum_{n \geq 1} y^n t^n = 1 + \frac{yt}{1 - yt}. \quad (2.44)$$

Now suppose $a \leq n - 2$, and put $m = n - a - 1 \geq 1$. Then

$$\sigma = 12 \cdots a n \tau^{+a}, \text{ with } \tau \in \mathfrak{S}_m(213, 231). \quad (2.45)$$

Since n occurs before all letters of τ^{+a} , the pair $(n - 1, n)$ always creates one new inverse descent. The remaining inverse descents are exactly those coming from τ . Hence,

$$\text{ides}(\sigma) = \text{ides}(\tau) + 1. \quad (2.46)$$

For the pixed statistic, we distinguish whether τ is a desarrangement. If τ is a desarrangement, then the suffix τ^{+a} is itself a desarrangement. By the pixed factorization of σ , we have $\text{pix}(\sigma) = a + 1$. If τ is not a desarrangement, then the prefix cannot include the letter n . In this case the suffix $n \tau^{+a}$ is a desarrangement. Then we have $\text{pix}(\sigma) = a$. Therefore

$$\text{pix}(\sigma) = \begin{cases} a + 1, & \text{if } \tau \text{ is a desarrangement;} \\ a, & \text{otherwise.} \end{cases} \quad (2.47)$$

Define the following two enumerative polynomials

$$A(t; x) = \sum_{n \geq 0} \left(\sum_{\sigma \in \mathfrak{S}_n(213, 231)} x^{\text{ides}(\sigma)} \right) t^n, \quad (2.48)$$

and

$$B(t; x) = \sum_{n \geq 1} \left(\sum_{\substack{\sigma \in \mathfrak{S}_n(213, 231) \\ \sigma \text{ is a desarrangement}}} x^{\text{ides}(\sigma)} \right) t^n. \quad (2.49)$$

We first compute $A(t; x)$. Together with the empty permutation, identity permutations contribute

$$1 + \sum_{n \geq 1} t^n = \frac{1}{1 - t}.$$

For the non-identity part, using the decomposition (2.45), the initial segment $12 \cdots a$ contributes t^a , the letter n contributes t , and by (2.46) it also contributes a factor x . The remaining nonempty permutation τ contributes $A(t; x) - 1$. Thus

$$A(t; x) = \frac{1}{1-t} + \frac{xt}{1-t} (A(t; x) - 1). \quad (2.50)$$

Solving this equation gives

$$A(t; x) = \frac{1 - xt}{1 - (1+x)t}. \quad (2.51)$$

Next we compute $B(t; x)$. A desarrangement in $\mathfrak{S}_n(213, 231)$ cannot begin with a nonempty increasing segment $12 \cdots a$ with $a \geq 1$, since then the first ascent occurs at position 1. Hence, in the decomposition (2.41), we must have $a = 0$. Thus every nonempty desarrangement in this class is of the form

$$\sigma = n\tau, \text{ with } \tau \in \mathfrak{S}_{n-1}(213, 231).$$

Moreover, τ is nonempty, since the singleton permutation is not a desarrangement.

The first ascent of $n\tau$ occurs one position later than the first ascent of τ . Thus the parity of the position of the first ascent is reversed. Consequently, $n\tau$ is a desarrangement if and only if τ is not a desarrangement. Since the initial letter n creates one new inverse descent, we obtain

$$B(t; x) = xt(A(t; x) - 1 - B(t; x)). \quad (2.52)$$

Therefore

$$B(t; x) = \frac{xt(A(t; x) - 1)}{1 + xt}. \quad (2.53)$$

Substituting $A(t; x)$ by (2.51), we obtain

$$B(t; x) = \frac{xt^2}{(1+xt)(1-(1+x)t)}. \quad (2.54)$$

We now return to $P_{T_2}(t; x, y)$. The contribution of empty permutation and identity permutations is already given by (2.44). For the non-identity part, from the decomposition $\sigma = 12 \cdots a n \tau^{+a}$, with $\tau \in \mathfrak{S}_m(213, 231)$, we see that the initial segment $12 \cdots a$ contributes $(yt)^a$, and summing over all $a \geq 0$ gives

$$\sum_{a \geq 0} (yt)^a = \frac{1}{1-yt}. \quad (2.55)$$

The letter n contributes t and creates one new inverse descent, hence gives a factor xt . Thus the common contribution is

$$\frac{xt}{1-yt}. \quad (2.56)$$

It remains to account for the contribution of τ . All nonempty τ 's contribute $A(t; x) - 1$. However, by (2.47), when τ is a desarrangement, the value of $\text{pix}(\sigma)$ is $a + 1$ rather than a , so these terms acquire one additional factor y . Therefore the contribution of the τ -part is

$$A(t; x) - 1 + (y - 1)B(t; x). \quad (2.57)$$

From the above argument, we see that the contribution of non-identity permutations is

$$\frac{xt}{1-yt} (A(t; x) - 1 + (y - 1)B(t; x)). \quad (2.58)$$

It follows that

$$P_{T_2}(t; x, y) = 1 + \frac{yt}{1-yt} + \frac{xt}{1-yt} (A(t; x) - 1 + (y-1)B(t; x)). \quad (2.59)$$

Substituting $A(t; x)$ and $B(t; x)$ by (2.51) and (2.53), respectively, we obtain (2.43). $\square$

2.1.3. *The class $\mathfrak{S}_n(132, 321)$.* By the Simion–Schmidt structural description [SS85, Proposition 11], for some $1 \leq k \leq m \leq n$, every permutation $\sigma_{k,m}$ in $\mathfrak{S}_n(132, 321)$ has the form

$$\sigma_{k,m} := (m-k+1)(m-k+2) \cdots m 12 \cdots (m-k)(m+1) \cdots n. \quad (2.60)$$

Note that in the case $m = k = n$, $\sigma_{n,n}$ is the identity permutation.

Theorem 2.7. *Let $T_3 = \{132, 321\}$. We have*

$$F_{T_3}(t; x, y) = \frac{1}{1-yt} + \frac{xt^2}{(1-t)^2(1-yt)}. \quad (2.61)$$

Proof. The identity permutation $\sigma_{n,n}$ contributes y^n to the coefficient of t^n in $F_{T_3}(t; x, y)$. For a non-identity element $\sigma_{k,m}$, by (2.60), we see that the three displayed segments are increasing, i.e., $(m-k+1, \dots, m)$, $(1, \dots, m-k)$ and $(m+1, \dots, n)$, and the only descent occurs between the first and second segments. Thus, we have

$$\text{des}(\sigma_{k,m}) = 1. \quad (2.62)$$

The fixed points are precisely those in the final segment $m+1, m+2, \dots, n$, and hence

$$\text{fix}(\sigma_{k,m}) = n - m. \quad (2.63)$$

For a fixed m , there are exactly $m-1$ non-identity permutations, and each permutation contributes xy^{n-m} . Thus,

$$\sum_{\sigma \in \mathfrak{S}_n(132, 321)} x^{\text{des}(\sigma)} y^{\text{fix}(\sigma)} = y^n + x \sum_{m=2}^n (m-1) y^{n-m}. \quad (2.64)$$

Together with the contribution 1 of the empty permutation, multiplying (2.64) by t^n and summing over $n \geq 1$ yields (2.61). $\square$

Theorem 2.8. *Let $T_3 = \{132, 321\}$. We have*

$$P_{T_3}(t; x, y) = \frac{1}{1-yt} + \frac{xt^2}{(1-t)^2(1-yt)}. \quad (2.65)$$

Proof. For the identity permutation $\sigma_{n,n}$, we have $\text{idex}(\sigma_{n,n}) = 0$. By (2.60), it is easy to see that, for $1 \leq k < m \leq n$

$$\sigma_{k,m}^{-1} = (k+1)(k+2) \cdots m 12 \cdots k (m+1) \cdots n = \sigma_{m-k,m}. \quad (2.66)$$

It follows that $\text{idex}(\sigma_{k,m}) = 1$. The length of the maximal consecutive increasing prefix of $\sigma_{k,m}$ is equal to k . In order to make the suffix of $\sigma_{k,m}$ a desarrangement, we need to have the following form

$$m, 1, 2, \dots, m-k, m+1, \dots, n. \quad (2.67)$$

Hence, we have $\text{pix}(\sigma_{k,m}) = k-1$. It follows that

$$\sum_{\sigma \in \mathfrak{S}_n(132, 321)} x^{\text{idex}(\sigma)} y^{\text{pix}(\sigma)} = y^n + \sum_{m=2}^n \sum_{k=1}^{m-1} xy^{k-1} = y^n + \sum_{j=0}^{n-2} (n-1-j)y^j. \quad (2.68)$$

Together with the contribution 1 of the empty permutation, multiplying (2.68) by t^n and summing over $n \geq 1$ yields (2.65). $\square$

2.2. Triple-avoidance classes in $\mathcal{P}$. In this section, we prove Theorem 1.2 for the remaining three-avoiding classes Π , namely $\{123, 132, 312\}$, $\{123, 213, 231\}$, $\{132, 312, 321\}$, $\{213, 231, 321\}$, $\{213, 231, 312\}$, and $\{123, 312, 321\}$ by computing their generating functions $F_\Pi(t; x, y)$ and $P_\Pi(t; x, y)$, respectively.

2.2.1. The class $\mathfrak{S}_n(123, 132, 312)$. Following the Simion-Schmidt description [SS85, Proposition 16], the permutations in $\mathfrak{S}_n(123, 132, 312)$ have the form

$$\hat{\sigma}_{n,k} = (n-1)(n-2) \cdots k n (k-1)(k-2) \cdots 1, \text{ for } 1 \leq k \leq n. \quad (2.69)$$

Theorem 2.9. *Let $T_4 = \{123, 132, 312\}$. We have*

$$F_{T_4}(t; x, y) = 1 + yt + t^2 \left(\frac{y^2 + x}{1 - x^2 t^2} + \frac{(1+y)x^2 t^2}{(1 - x^2 t^2)^2} \right) + t \left(\frac{(1+y)xt^2}{(1 - x^2 t^2)^2} + \frac{yx^2 t^2}{1 - x^2 t^2} \right). \quad (2.70)$$

Proof. By (2.69), we see that

$$\sum_{\sigma \in \mathfrak{S}_n(T_4)} x^{\text{des}(\sigma)} y^{\text{fix}(\sigma)} = \sum_{k=1}^n x^{\text{des}(\hat{\sigma}_{n,k})} y^{\text{fix}(\hat{\sigma}_{n,k})}. \quad (2.71)$$

Let us first determine the number of descents of $\hat{\sigma}_{n,k}$. If $k < n$, then the only ascent occurs at position $n - k$, while all other positions $i \in [n - 1] \setminus \{n - k\}$ are descents. If $k = n$, then $\hat{\sigma}_{n,n} = n(n-1) \cdots 1$, so every position $i \in [n - 1]$ is a descent. Therefore,

$$\text{des}(\hat{\sigma}_{n,k}) = \begin{cases} n - 2, & \text{if } 1 \leq k < n; \\ n - 1, & \text{if } k = n. \end{cases} \quad (2.72)$$

We next compute the number of fixed points of $\hat{\sigma}_{n,k}$. From (2.69), the letter n occurs in position $n - k + 1$. Hence, it is easy to see that the letter n is a fixed point if and only if $k = 1$. For $k \leq a \leq n - 1$, the letter a occurs in position $n - a$, and it is a fixed point if and only if n is even and $a = n/2$. For $1 \leq a \leq k - 1$, the letter a occurs in position $n - a + 1$, and it is a fixed point if and only if n is odd and $a = (n + 1)/2$. It follows that

$$\text{fix}(\hat{\sigma}_{n,k}) = \chi(k = 1) + \chi(n \text{ is even and } k \leq n/2) + \chi(n \text{ is odd and } k > (n + 1)/2). \quad (2.73)$$

Equivalently, if $n = 2m + 1$, then

$$\text{fix}(\hat{\sigma}_{2m+1,k}) = \begin{cases} 1, & k = 1 \text{ or } m + 2 \leq k \leq 2m + 1; \\ 0, & 2 \leq k \leq m + 1, \end{cases} \quad (2.74)$$

while if $n = 2m$, then

$$\text{fix}(\hat{\sigma}_{2m,k}) = \begin{cases} 2, & k = 1; \\ 1, & 2 \leq k \leq m; \\ 0, & m + 1 \leq k \leq 2m. \end{cases} \quad (2.75)$$

Hence, for $n = 2m + 1$, by (2.74), we have

$$\sum_{\sigma \in \mathfrak{S}_{2m+1}(T_4)} x^{\text{des}(\sigma)} y^{\text{fix}(\sigma)} = x^{2m} y + x^{2m-1} m(1 + y). \quad (2.76)$$

For $n = 2m$, by (2.75), we have

$$\sum_{\sigma \in \mathfrak{S}_{2m}(T_4)} x^{\text{des}(\sigma)} y^{\text{fix}(\sigma)} = x^{2m-1} + x^{2m-2}((m-1)(1+y) + y^2). \quad (2.77)$$

Together with the initial contributions 1 and yt corresponding to $n = 0$ and $n = 1$, respectively, the stated generating function (2.70) follows by multiplying (2.76) and (2.77) by t^{2m+1} and t^{2m} , respectively, then summing over $m \geq 1$. $\square$

Theorem 2.10. *Let $T_4 = \{123, 132, 312\}$. We have*

$$P_{T_4}(t; x, y) = 1 + yt + t^2 \left(\frac{y^2 + x}{1 - x^2 t^2} + \frac{(1+y)x^2 t^2}{(1 - x^2 t^2)^2} \right) + t \left(\frac{(1+y)xt^2}{(1 - x^2 t^2)^2} + \frac{yx^2 t^2}{1 - x^2 t^2} \right). \quad (2.78)$$

Proof. First, let us determine ides . From (2.69), a direct check gives

$$\text{ides}(\hat{\sigma}_{n,k}) = \begin{cases} n-2, & \text{if } 1 \leq k < n; \\ n-1, & \text{if } k = n. \end{cases} \quad (2.79)$$

Now we compute pix . Recall that $\text{pix}(\sigma)$ is the length of the initial increasing word in the pixed factorization of σ . If $k < n$, then the unique ascent of $\hat{\sigma}_{n,k}$ occurs at position $n-k$. Therefore $\hat{\sigma}_{n,k}$ itself is a desarrangement precisely when $n-k$ is even.

Suppose first that $n = 2m+1$. Then $n-k$ is even exactly when k is odd. Thus, for odd $k < 2m+1$, we have $\text{pix}(\hat{\sigma}_{2m+1,k}) = 0$. For even k , deleting the first letter shifts the unique ascent to an even position, so the remaining word is a desarrangement and $\text{pix}(\hat{\sigma}_{2m+1,k}) = 1$. Finally, when $k = 2m+1$, deleting the first letter leaves a decreasing word of even length, which is a desarrangement. Hence

$$\text{pix}(\hat{\sigma}_{2m+1,k}) = \begin{cases} 0, & k \text{ is odd and } k < 2m+1; \\ 1, & k \text{ is even}; \\ 1, & k = 2m+1. \end{cases} \quad (2.80)$$

Now suppose that $n = 2m$. Then $n-k$ is even exactly when k is even, so $\text{pix}(\hat{\sigma}_{2m,k}) = 0$ for even k . If k is odd and $k < 2m-1$, deleting the first letter shifts the unique ascent to an even position, and hence $\text{pix}(\hat{\sigma}_{2m,k}) = 1$. In the remaining case $k = 2m-1$, the first two letters form the increasing word $(2m-1, 2m)$, and deleting them leaves a decreasing word of even length, which is a desarrangement. If $n = 2m$, then

$$\text{pix}(\hat{\sigma}_{2m,k}) = \begin{cases} 0, & k \text{ is even}; \\ 1, & k \text{ is odd and } k < 2m-1; \\ 2, & k = 2m-1. \end{cases} \quad (2.81)$$

Therefore, using (2.79) and (2.80), for $n = 2m+1$,

$$\sum_{\sigma \in \mathfrak{S}_{2m+1}(T_4)} x^{\text{ides}(\sigma)} y^{\text{pix}(\sigma)} = x^{2m} y + x^{2m-1} m(1+y), \quad (2.82)$$

and using (2.79) and (2.81), for $n = 2m$,

$$\sum_{\sigma \in \mathfrak{S}_{2m}(T_4)} x^{\text{ides}(\sigma)} y^{\text{pix}(\sigma)} = x^{2m-1} + x^{2m-2}((m-1)(1+y) + y^2). \quad (2.83)$$

Together with the initial contributions 1 and yt corresponding to $n = 0$ and $n = 1$, respectively, the stated generating function (2.78) follows by multiplying (2.82) and (2.83) by t^{2m+1} and t^{2m} , respectively, then summing over $m \geq 1$. $\square$

2.2.2. The class $\mathfrak{S}_n(123, 213, 231)$. Since ρ is a bijection between $\mathfrak{S}_n(123, 132, 312)$ and $\mathfrak{S}_n(123, 213, 231)$, applying ρ to (2.69) shows that every permutation in $\mathfrak{S}_n(123, 213, 231)$ has the form

$$\tilde{\sigma}_{n,k} = n(n-1) \cdots (n-k+2) 1 (n-k+1)(n-k) \cdots 2, \quad \text{for } 1 \leq k \leq n. \quad (2.84)$$

Theorem 2.11. *Let $T_5 = \{123, 213, 231\}$. We have*

$$F_{T_5}(t; x, y) = 1 + yt + t^2 \left(\frac{y^2 + x}{1 - x^2 t^2} + \frac{(1+y)x^2 t^2}{(1 - x^2 t^2)^2} \right) + t \left(\frac{(1+y)xt^2}{(1 - x^2 t^2)^2} + \frac{yx^2 t^2}{1 - x^2 t^2} \right). \quad (2.85)$$

Proof. By Lemma 2.1, the reversal-complement map $\sigma \mapsto \sigma^{rc}$ restricts a bijection between $\mathfrak{S}_n(123, 213, 231)$ and $\mathfrak{S}_n(123, 132, 312)$, thus we have $F_{T_5}(t; x, y) = F_{T_4}(t; x, y)$, where $T_4 = \{123, 132, 312\}$. Then the stated formula follows immediately from Theorem 2.9. $\square$

Theorem 2.12. *Let $T_5 = \{123, 213, 231\}$. We have*

$$P_{T_5}(t; x, y) = 1 + yt + t^2 \left(\frac{y^2 + x}{1 - x^2 t^2} + \frac{(1+y)x^2 t^2}{(1 - x^2 t^2)^2} \right) + t \left(\frac{(1+y)xt^2}{(1 - x^2 t^2)^2} + \frac{yx^2 t^2}{1 - x^2 t^2} \right). \quad (2.86)$$

Proof. It is straightforward to see that

$$\text{ides}(\tilde{\sigma}_{n,k}) = \begin{cases} n-2, & 1 \leq k < n; \\ n-1, & k = n. \end{cases} \quad (2.87)$$

Next we compute $\text{pix}(\tilde{\sigma}_{n,k})$. From the explicit form of $\tilde{\sigma}_{n,k}$, if $k < n$, the unique ascent occurs at position k , namely the adjacent pair $1 < n - k + 1$. Thus $\tilde{\sigma}_{n,k}$ is itself a desarrangement exactly when k is even. If k is odd and $k < n$, deleting the first letter shifts this unique ascent to position $k - 1$, which is even.

Now suppose $n = 2m + 1$. If k is even, then $\tilde{\sigma}_{2m+1,k}$ is already a desarrangement, so its pix is 0. If k is odd and $k < 2m + 1$, deleting the first letter gives a desarrangement, so the pix is 1. Finally, when $k = 2m + 1$, the word $\tilde{\sigma}_{2m+1,2m+1} = (2m+1)(2m) \cdots 1$ is decreasing of odd length; deleting the first letter leaves a decreasing word of even length, which is a desarrangement. Hence

$$\text{pix}(\tilde{\sigma}_{2m+1,k}) = \begin{cases} 0, & k \text{ is even;} \\ 1, & k \text{ is odd.} \end{cases} \quad (2.88)$$

Next suppose $n = 2m$. If k is even, then $\tilde{\sigma}_{2m,k}$ is already a desarrangement, so its pix is 0. If k is odd and $3 \leq k \leq 2m - 1$, deleting the first letter shifts the unique ascent to the even position $k - 1$, so the pix is 1. The remaining case is $k = 1$. In this case $\tilde{\sigma}_{2m,1} = 1 \ 2m \ 2m - 1 \ \dots \ 2$ and the initial increasing word has length 2, since deleting the first two letters leaves the decreasing word $(2m - 1) \cdots 2$ of even length, which is a desarrangement. Therefore

$$\text{pix}(\tilde{\sigma}_{2m,k}) = \begin{cases} 0, & k \text{ is even;} \\ 1, & k \text{ is odd and } 3 \leq k \leq 2m - 1; \\ 2, & k = 1. \end{cases} \quad (2.89)$$

Combining (2.87) and (2.88), for $n = 2m + 1$, we get

$$\sum_{\sigma \in \mathfrak{S}_{2m+1}(T_5)} x^{\text{idex}(\sigma)} y^{\text{pix}(\sigma)} = x^{2m} y + m x^{2m-1} (1 + y). \quad (2.90)$$

Similarly, combining (2.87) and (2.89), for $n = 2m$, we get

$$\sum_{\sigma \in \mathfrak{S}_{2m}(T_5)} x^{\text{idex}(\sigma)} y^{\text{pix}(\sigma)} = x^{2m-1} + x^{2m-2} ((m-1)(1+y) + y^2). \quad (2.91)$$

Together with the initial contributions 1 and yt corresponding to $n = 0$ and $n = 1$, respectively, the stated generating function follows by multiplying (2.90) and (2.91) by t^{2m+1} and t^{2m} , respectively, and summing over $m \geq 1$. $\square$

2.2.3. The class $\mathfrak{S}_n(132, 312, 321)$. We claim that every permutation in $\mathfrak{S}_n(132, 312, 321)$ has the following form

$$\bar{\sigma}_{n,k} = 23 \cdots k 1 (k+1)(k+2) \cdots n, \text{ for } 1 \leq k \leq n. \quad (2.92)$$

Here, the case $k = 1$ gives the identity permutation $\bar{\sigma}_{n,1} = 12 \cdots n$.

Indeed, let $\sigma \in \mathfrak{S}_n(132, 312, 321)$, and let $j = \sigma^{-1}(1)$ be the position of 1. Since σ avoids 321, there are no descents among the letters lying to the left of 1. Hence, the subword to the left of 1 is increasing. Similarly, since σ avoids 132, the subword to the right of 1 is also increasing.

It remains to determine which letters occur on each side of 1. Suppose that there exist letters a to the left of 1 and b to the right of 1 with $a > b$. Then the three letters $a 1 b$, in this order, forms a 312-pattern, contradicting the avoiding condition. Therefore, every letter to the left of 1 is smaller than every letter to the right of 1. It follows that the letters to the left of 1 are precisely $2, 3, \dots, k$ for some k , and the letters to the right of 1 are $k+1, k+2, \dots, n$, that is (2.92).

Theorem 2.13. *Let $T_6 = \{132, 312, 321\}$. We have*

$$F_{T_6}(t; x, y) = 1 + \frac{yt}{1-yt} + \frac{xt^2}{(1-t)(1-yt)}. \quad (2.93)$$

Proof. From the characterization (2.92), for $k = 1$, we have $\bar{\sigma}_{n,1} = 12 \cdots n$. Hence

$$\text{des}(\bar{\sigma}_{n,1}) = 0 \text{ and } \text{fix}(\bar{\sigma}_{n,1}) = n. \quad (2.94)$$

For $2 \leq k \leq n$, it is straightforward to see that the permutation $\bar{\sigma}_{n,k}$ has exactly one descent, and its fixed points are exactly $k+1, k+2, \dots, n$, i.e.,

$$\text{des}(\bar{\sigma}_{n,k}) = 1 \text{ and } \text{fix}(\bar{\sigma}_{n,k}) = n - k. \quad (2.95)$$

Therefore, combining (2.94) and (2.95), we obtain

$$\sum_{\sigma \in \mathfrak{S}_n(T_6)} x^{\text{des}(\sigma)} y^{\text{fix}(\sigma)} = y^n + \sum_{k=2}^n x y^{n-k}. \quad (2.96)$$

Consequently, after multiplying by t^n and summing over $n \geq 1$, and after adding the initial contribution 1 of the empty permutation corresponding to $n = 0$, we obtain (2.93). $\square$

Theorem 2.14. *Let $T_6 = \{132, 312, 321\}$. We have*

$$P_{T_6}(t; x, y) = 1 + \frac{yt}{1-yt} + \frac{xt^2}{(1-t)(1-yt)}. \quad (2.97)$$

Proof. From the characterization (2.92), for $k = 1$, we have $\bar{\sigma}_{n,1} = 12 \cdots n$. Hence

$$\text{ides}(\bar{\sigma}_{n,1}) = 0 \text{ and } \text{pix}(\bar{\sigma}_{n,1}) = n. \quad (2.98)$$

It remains to determine $\text{pix}(\bar{\sigma}_{n,r})$. Recall that $\text{pix}(\bar{\sigma})$ is the length of the increasing prefix in the fixed factorization of $\bar{\sigma}$. For $2 \leq k \leq n$, the maximal increasing prefix of $\bar{\sigma}_{n,k}$ which occurs before the desarrangement is $23 \cdots (k-1)$, that is,

$$\bar{\sigma}_{n,k} = \underbrace{2 \cdots (k-1)}_{\iota} \underbrace{k1(k+1) \cdots n}_{\delta}.$$

Thus, we have

$$\text{pix}(\bar{\sigma}_{n,k}) = k - 2. \quad (2.99)$$

Therefore, combining (2.98) and (2.99), we have

$$\sum_{\sigma \in \mathfrak{S}_n(T_6)} x^{\text{ides}(\sigma)} y^{\text{pix}(\sigma)} = y^n + \sum_{k=2}^n x y^{n-k}. \quad (2.100)$$

Consequently, after multiplying by t^n and summing over $n \geq 1$, and after adding the initial contribution 1 of the empty permutation corresponding to $n = 0$, we obtain (2.97). $\square$

2.2.4. The class $\mathfrak{S}_n(213, 231, 321)$. Since the reversal-complement map $\sigma \mapsto \sigma^{rc}$ is a bijection between $\mathfrak{S}_n(132, 312, 321)$ and $\mathfrak{S}_n(213, 231, 321)$, by the form of permutations in $\mathfrak{S}_n(132, 312, 321)$ (see (2.92)), we derive that every permutation in $\mathfrak{S}_n(213, 231, 321)$ has the following form

$${}^*\sigma_{n,k} := 12 \cdots (k-1) n k (k+1) \cdots (n-1), \text{ for } 1 \leq k \leq n. \quad (2.101)$$

Theorem 2.15. *Let $T_7 = \{213, 231, 321\}$. We have*

$$F_{T_7}(t; x, y) = 1 + \frac{yt}{1-yt} + \frac{xt^2}{(1-t)(1-yt)}. \quad (2.102)$$

Proof. By Lemma 2.1, the reversal-complement map $\sigma \mapsto \sigma^{rc}$ restricts a bijection between $\mathfrak{S}_n(213, 231, 321)$ and $\mathfrak{S}_n(132, 312, 321)$, thus we have $F_{T_7}(t; x, y) = F_{T_6}(t; x, y)$, where $T_6 = \{132, 312, 321\}$. By the formula for $F_{T_6}(t; x, y)$ given in (2.93), we obtain (2.102). $\square$

Theorem 2.16. *Let $T_7 = \{213, 231, 321\}$. We have*

$$P_{T_7}(t; x, y) = 1 + \frac{yt}{1-yt} + \frac{xt^2}{(1-t)(1-yt)}. \quad (2.103)$$

Proof. From the characterization (2.101), for $k = n$, we have ${}^*\sigma_{n,n} = 12 \cdots n$. Hence,

$$\text{ides}({}^*\sigma_{n,n}) = 0 \text{ and } \text{pix}({}^*\sigma_{n,n}) = n. \quad (2.104)$$

For $1 \leq k \leq n-1$, one has

$$\text{ides}({}^*\sigma_{n,k}) = 1 \quad (2.105)$$

Recall that $\text{pix}(\pi)$ is the length of the initial increasing factor in the fixed factorization of π . For $1 \leq k \leq n-1$, we may write

$${}^*\sigma_{n,k} = \underbrace{12 \cdots (k-1)}_{\iota} \underbrace{nk(k+1) \cdots (n-1)}_{\delta}.$$

The word $\iota = 12 \cdots (k-1)$ is increasing. Moreover, the suffix $\delta = nk(k+1) \cdots (n-1)$ is a desarrangement: if $k \leq n-2$, then its first ascent occurs at the second position, while if

$k = n - 1$, then $\delta = n(n - 1)$ has no ascent and is a desarrangement by convention. Thus the above decomposition is precisely the pixed factorization of ${}^*\sigma_{n,k}$. Consequently,

$$\text{pix}({}^*\sigma_{n,k}) = k - 1. \quad (2.106)$$

Combining (2.104), (2.105) and (2.106), we obtain, for $n \geq 1$,

$$\sum_{\sigma \in \mathfrak{S}_n(T_7)} x^{\text{idex}(\sigma)} y^{\text{pix}(\sigma)} = y^n + \sum_{k=1}^{n-1} xy^{k-1}. \quad (2.107)$$

Consequently, after multiplying by t^n and summing over $n \geq 1$, and after adding the initial contribution 1 of the empty permutation corresponding to $n = 0$, we obtain (2.103). $\square$

2.2.5. The class $\mathfrak{S}_n(213, 231, 312)$. We claim that every permutation in $\mathfrak{S}_n(213, 231, 312)$ is of the form

$$\dot{\sigma}_{n,k} = 12 \cdots k n(n-1) \cdots (k+1), \text{ for } 0 \leq k \leq n-1. \quad (2.108)$$

Indeed, let $\sigma \in \mathfrak{S}_n(213, 231, 312)$, and consider the position of the letter 1. Since 1 is the smallest letter, if there were letters on both sides of 1, say a to its left and b to its right, then the subsequence $a1b$ would form a 213-pattern if $a < b$, and a 312-pattern if $a > b$. Thus the letter 1 must occur at one of the two ends of σ .

If 1 is the last letter, then the subword preceding it must be decreasing. Indeed, any increasing pair $a < b$ occurring before 1 would give a 231-pattern $ab1$. Hence, in this case,

$$\sigma = n(n-1) \cdots 21,$$

which corresponds to $k = 0$ in (2.108).

Otherwise, 1 must be the first letter. Deleting this initial 1 and standardizing the remaining word, we obtain a permutation of $\mathfrak{S}_{n-1}(213, 231, 312)$. Repeating the same argument, we find that the letters $1, 2, \dots, k$ must appear successively at the beginning of σ , until the first step at which the smallest remaining letter is placed at the right end. At that point, the remaining letters must appear in decreasing order. Consequently, for some $0 \leq k \leq n-1$,

$$\sigma = \dot{\sigma}_{n,k} = 12 \cdots k n(n-1) \cdots (k+1),$$

as claimed.

Theorem 2.17. *Let $T_8 = \{213, 231, 312\}$. We have*

$$F_{T_8}(t; x, y) = 1 + \frac{yt + xt^2}{(1 - yt)(1 - x^2t^2)}. \quad (2.109)$$

Proof. From (2.108), the initial segment $12 \cdots k$ is increasing, whereas the final segment $n(n-1) \cdots (k+1)$ is decreasing and has length $n - k$. Hence the descents occur precisely inside this final decreasing segment. Therefore

$$\text{des}(\dot{\sigma}_{n,k}) = n - k - 1. \quad (2.110)$$

It remains to determine the number of fixed points. The letters $1, 2, \dots, k$ are fixed points, and hence contribute k fixed points. Now consider the final decreasing segment. Its positions are $k+1, k+2, \dots, n$, and its corresponding values are $n, n-1, \dots, k+1$. The j -th entry of this segment, where $1 \leq j \leq n - k$, lies in position $k+j$ and has value $n - j + 1$. It is fixed if and only if $k+j = n - j + 1$, or equivalently, $2j = n - k + 1$. Thus the final decreasing segment contributes exactly one additional fixed point when $n - k$ is odd. Consequently,

$$\text{fix}(\dot{\sigma}_{n,k}) = k + \chi(n - k \text{ is odd}). \quad (2.111)$$

Combining (2.110) and (2.111), we obtain, for $n \geq 1$,

$$\sum_{\sigma \in \mathfrak{S}_n(T_8)} x^{\text{des}(\sigma)} y^{\text{fix}(\sigma)} = \sum_{k=0}^{n-1} x^{n-k-1} y^{k+\chi(n-k \text{ is odd})}. \quad (2.112)$$

Taking also into account the empty permutation for $n = 0$, which contributes 1, and multiplying both sides of (2.112) by t^n and summing over $n \geq 1$, we get

$$\begin{aligned} F_{T_8}(t; x, y) &= 1 + \sum_{n \geq 1} t^n \sum_{\sigma \in \mathfrak{S}_n(T_8)} x^{\text{des}(\sigma)} y^{\text{fix}(\sigma)} \\ &= 1 + \sum_{n \geq 1} \sum_{k=0}^{n-1} t^n x^{n-k-1} y^{k+\chi(n-k \text{ is odd})}. \end{aligned}$$

Setting $\ell = n - k$, the length of the final decreasing segment in the above identity. Then for $\ell \geq 1$, $k \geq 0$ and $n = k + \ell$, we have

$$\begin{aligned} F_{T_8}(t; x, y) &= 1 + \sum_{k \geq 0} \sum_{\ell \geq 1} t^{k+\ell} x^{\ell-1} y^{k+\chi(\ell \text{ is odd})} \\ &= 1 + \left(\sum_{k \geq 0} (yt)^k \right) \left(\sum_{\ell \geq 1} t^\ell x^{\ell-1} y^{\chi(\ell \text{ is odd})} \right) \\ &= 1 + \frac{1}{1 - yt} \sum_{\ell \geq 1} t^\ell x^{\ell-1} y^{\chi(\ell \text{ is odd})}. \end{aligned} \quad (2.113)$$

Splitting according to the parity of ℓ , the last sum of (2.113) is equal to

$$\sum_{\ell \geq 1} t^\ell x^{\ell-1} y^{\chi(\ell \text{ is odd})} = \sum_{m \geq 0} t^{2m+1} x^{2m} y + \sum_{m \geq 1} t^{2m} x^{2m-1} = \frac{yt + xt^2}{1 - x^2 t^2}. \quad (2.114)$$

Combining this and (2.113) yields (2.109). $\square$

Theorem 2.18. *Let $T_8 = \mathfrak{S}_n(213, 231, 312)$. We have*

$$P_{T_8}(t; x, y) = 1 + \frac{yt + xt^2}{(1 - yt)(1 - x^2 t^2)}. \quad (2.115)$$

Proof. Recall that $\text{ides}(\sigma)$ counts the values $v \in [n - 1]$ such that $v + 1$ appears before v in σ . By (2.108), we see that each v is an inverse descent of $\hat{\sigma}_{n,k}$ for $k + 1 \leq v \leq n - 1$. Since the final segment is decreasing, while the initial segment $12 \cdots k$ is increasing and k appears before $k + 1$. Hence, k is also an inverse descent of $\hat{\sigma}_{n,k}$.

$$\text{ides}(\hat{\sigma}_{n,k}) = n - k - 1. \quad (2.116)$$

It remains to determine the number of fixed points. For $\hat{\sigma}_{n,k}$, the initial increasing part is $12 \cdots k$, and the remaining suffix is the decreasing word $n(n - 1) \cdots (k + 1)$, of length $n - k$. We distinguish two cases according to the parity of $n - k$. If $n - k$ is even, then the decreasing suffix $n(n - 1) \cdots (k + 1)$ is a derangement. Thus the fixed factorization is

$$\hat{\sigma}_{n,k} = \underbrace{12 \cdots k}_\iota \underbrace{n(n - 1) \cdots (k + 1)}_\delta,$$

and consequently $\text{pix}(\hat{\sigma}_{n,k}) = k$.

If $n - k$ is odd, then the decreasing suffix $n(n - 1) \cdots (k + 1)$ is not a desarrangement. In this case, the first letter n of the final segment must be included in the increasing prefix. The remaining suffix $(n - 1)(n - 2) \cdots (k + 1)$ has even length and is therefore a desarrangement. Hence the pixed factorization is

$$\circ\sigma_{n,k} = \underbrace{12 \cdots k n}_{\iota} \underbrace{(n - 1)(n - 2) \cdots (k + 1)}_{\delta},$$

and so $\text{pix}(\circ\sigma_{n,k}) = k + 1$. Thus, in all cases, we have

$$\text{pix}(\circ\sigma_{n,k}) = k + \chi(n - k \text{ is odd}). \quad (2.117)$$

Combining (2.116) and (2.117), we obtain, for fixed $n \geq 1$,

$$\sum_{\sigma \in \mathfrak{S}_n(T_8)} x^{\text{idex}(\sigma)} y^{\text{pix}(\sigma)} = \sum_{k=0}^{n-1} x^{n-k-1} y^{k+\chi(n-k \text{ is odd})}. \quad (2.118)$$

Since the right-hand side of (2.118) is identical to that of (2.112), the same summation argument as above gives $F_{T_8}(t; x, y) = P_{T_8}(t; x, y)$. Hence, by (2.109), we obtain the desired formula. $\square$

2.2.6. The class $\mathfrak{S}_n(123, 312, 321)$. Let $T_9 = \{123, 312, 321\}$. By the Erds–Szekeres theorem [Coh78, p.160], $\mathfrak{S}_n(123, 321) = \emptyset$ for $n \geq 5$, and therefore $\mathfrak{S}_n(T_9) = \emptyset$ for $n \geq 5$. For $1 \leq n \leq 4$, direct enumeration yields $\mathfrak{S}_1(T_9) = \{1\}$, $\mathfrak{S}_2(T_9) = \{12, 21\}$, $\mathfrak{S}_3(T_9) = \{132, 213, 231\}$, and $\mathfrak{S}_4(T_9) = \{2143\}$. Consequently, we have

$$F_{T_9}(t; x, y) = P_{T_9}(t; x, y) = 1 + yt + (y^2 + x)t^2 + x(2y + 1)t^3 + x^2t^4.$$

3. CONCLUDING REMARKS

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